

Automatic Rigidity for Collatz-Invariant 2-Automatic Sequences

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Abstract

Let $a = (a_n)_{n \geq 0}$ be a sequence over a finite alphabet. We prove that if a is 2-automatic and satisfies

$$a_{2n} = a_n, \quad a_{2n+1} = a_{3n+2} \quad (n \geq 0),$$

then a is constant on the positive integers, while a_0 is unconstrained. Equivalently, every finite-alphabet 2-automatic coloring invariant under the shortcut Collatz map is trivial on the positive integers. The proof has two stages. First, every 3-kernel subsequence of the odd-indexed sequence $b_n = a_{2n+1}$ is embedded into the finite 2-kernel of a ; the arithmetic input is that 2 is a primitive root modulo 3^j for every $j \geq 1$. Thus b is simultaneously 2- and 3-automatic, and Cobham's theorem implies that b is ultimately periodic. Second, two identities inherited from Collatz invariance eliminate every nontrivial eventual period and force b , hence a on the positive integers, to be constant. As a consequence, the Collatz conjecture is equivalent to 2-automaticity of the characteristic sequence of the basin of 1. This equivalence is a reformulation, not a proof, of the Collatz conjecture.

Keywords: Collatz conjecture; automatic sequence; Cobham theorem; finite automata; k -kernel; invariant set.

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Author's Note: Origin of This Work

I am not a professional mathematician. One of the reasons I began exploring the Collatz conjecture was not only my interest in the problem itself, but also a simple question: how far has OpenAI's ChatGPT (GPT-5.6 Sol) actually progressed as of 2026?

Rather than merely asking the AI to "solve the Collatz conjecture," I wanted to see how far an extended dialogue could go if I supplied my own questions, intuitions, and ideas, while allowing the AI to develop the mathematical details.

In many stages of this exploration, I did not provide detailed instructions for advanced proofs. My replies were often as short as "let us investigate that," "let us continue in that direction," or simply "let us continue."

Nevertheless, through repeated dialogue, a research-like process gradually emerged. I raised questions and chose directions; ChatGPT expanded the ideas, compared them with known mathematics, searched for possible errors and counterexamples, examined the literature, proposed proofs, and eventually assisted in formalizing substantial parts of the argument in Lean 4.

I then asked ChatGPT to compare the accumulated results of these AI-assisted explorations with the known literature and to examine, as broadly as possible, the question: "What structural obstruction still remains?" The rigidity phenomenon studied in this paper emerged from that process.

Thus, this paper has a second purpose in addition to its mathematical content. It is also a record of an experiment: how far can a person without professional training in mathematics explore the neighborhood of an unsolved problem through sustained dialogue with a contemporary AI system?

My role was not to carry out every calculation or proof myself, but to ask questions, choose directions, challenge conclusions, request verification, and repeatedly say, “let us continue.”

And one of the things I wanted to discover was precisely how far ChatGPT could respond to that request to continue.

1 Introduction

The shortcut Collatz map $T : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ is defined by

$$T(2n) = n, \quad T(2n+1) = 3n+2 \quad (n \geq 0). \quad (1)$$

For positive integers this is the usual Collatz map with one division by 2 built into the odd step. For general background on the $3x+1$ problem, see Lagarias [11].

Let Δ be a finite alphabet. A coloring $a : \mathbb{N}_0 \rightarrow \Delta$ is T -invariant when

$$a_n = a_{T(n)} \quad (n \geq 0).$$

Using (1), this is equivalent to

$$a_{2n} = a_n, \quad a_{2n+1} = a_{3n+2} \quad (n \geq 0). \quad (2)$$

Relations of this type occur in analytic formulations of the Collatz problem; see Berg and Meinardus [3, 4]. Finite-state descriptions have also appeared in several inverse-orbit settings. Shallit and Wilson [12] treated predecessor families with a fixed number of odd Collatz steps, Colussi [7] described finite convergence classes in binary form, and Stérin [13] constructed regular expressions for bounded-budget ancestor sets. Stérin and Woods [14] showed that the Collatz process embeds a base-3 to base-2 conversion algorithm. More recently, Dhiman and Pandey [8, 9] proved finite-automata non-definability results for certain two-variable transition and reachability relations associated with generalized Collatz maps.

The question considered here is different and one-dimensional: can a finite coloring that is nonconstant on the positive integers and invariant along every shortcut Collatz edge be 2-automatic? The answer is no.

Theorem 1.1 (Automatic rigidity). *Let Δ be a finite alphabet and let $a = (a_n)_{n \geq 0} \in \Delta^{\mathbb{N}_0}$ be 2-automatic. Assume (2) for every $n \geq 0$. Then there exists $c \in \Delta$ such that*

$$a_n = c \quad (n \geq 1).$$

The value a_0 is arbitrary.

No assumption concerning the truth of the Collatz conjecture is used in Theorem 1.1. The theorem is an automatic-sequence rigidity statement about the map (1). In the functional graph of T , equation (2) says that the coloring is constant along every directed edge; the theorem asserts that finite-state base-2 structure does not permit different colors on positive invariant components.

The proof separates naturally into two mechanisms. Section 3 transfers the odd-indexed subsequence from base 2 to base 3 and invokes Cobham’s theorem. Section 4 then uses two further Collatz identities to destroy the periodic structure supplied by Cobham. Section 5 records consequences for invariant sets and the basin of 1, and Section 6 compares the result with earlier finite-state approaches.

2 Automatic sequences and an arithmetic lemma

For an integer $k \geq 2$, the k -kernel of a sequence $u = (u_n)_{n \geq 0}$ is

$$K_k(u) := \{(u_{k^e n + r})_{n \geq 0} : e \geq 0, 0 \leq r < k^e\}.$$

When referring to base- k digit representations, we use the standard most-significant-digit-first convention, with no leading zeros and with 0 represented by the one-digit word 0. A sequence over a finite alphabet is k -automatic if and only if its k -kernel is finite [1]. We use the sequence form of Cobham's theorem (see [1], with the original recognizability theorem in [6]): if $k, \ell \geq 2$ are multiplicatively independent and a finite-alphabet sequence is both k -automatic and ℓ -automatic, then it is ultimately periodic.

The following elementary number-theoretic fact is the arithmetic input needed below.

Lemma 2.1. *For every $j \geq 1$, the integer 2 is a primitive root modulo 3^j .*

Proof. We first show that

$$v_3(4^{3^t} - 1) = t + 1 \quad (t \geq 0). \quad (3)$$

The case $t = 0$ is immediate. Suppose $x \equiv 1 \pmod{3}$ and $v_3(x - 1) = r \geq 1$. Since

$$x^3 - 1 = (x - 1)(x^2 + x + 1)$$

and $v_3(x^2 + x + 1) = 1$, cubing increases the 3-adic valuation of $x - 1$ by exactly one. Induction gives (3).

For $j = 1$, the order of 2 modulo 3 is 2. Let $j \geq 2$. Equation (3) gives

$$2^{2 \cdot 3^{j-1}} \equiv 1 \pmod{3^j}, \quad 2^{2 \cdot 3^{j-2}} \not\equiv 1 \pmod{3^j}.$$

Reduction modulo 3 sends any congruence modulo 3^j to a congruence modulo 3. Hence the order of 2 modulo 3, which is 2, divides the order of 2 modulo 3^j . The latter order also divides

$$\varphi(3^j) = 2 \cdot 3^{j-1}.$$

Every even divisor of $2 \cdot 3^{j-1}$ has the form $2 \cdot 3^t$. If $t \leq j - 2$, then the order would divide $2 \cdot 3^{j-2}$, contradicting the second congruence above. Hence the order is $2 \cdot 3^{j-1} = \varphi(3^j)$. $\square$

3 From the 2-kernel to the 3-kernel

The next lemma is the main transfer step.

Lemma 3.1 (Power-of-3 affine subsequences lie in the 2-kernel). *Assume that a satisfies (2). Let $j \geq 1$ and let c satisfy*

$$0 < c < 3^j, \quad 3 \nmid c.$$

Then

$$(a_{3^j n + c})_{n \geq 0} \in K_2(a).$$

Proof. Because c is a unit modulo 3^j and 2 generates $(\mathbb{Z}/3^j\mathbb{Z})^\times$ by Lemma 2.1, there exists $s \geq 0$ such that

$$2^s c \equiv -1 \pmod{3^j}. \quad (4)$$

Write

$$2^s c + 1 = q 3^j$$

with $q \geq 1$, and set $X_n = 3^j n + c$. Repeated use of the first identity in (2) gives

$$a_{X_n} = a_{2^s X_n}.$$

Whenever $y \equiv 2 \pmod{3}$, the second identity in (2) may be read backwards as

$$a_y = a_{(2y-1)/3}. \quad (5)$$

Starting with $2^s X_n$ and applying (5) exactly t times gives, for $0 \leq t \leq j$, the index

$$Y_t(n) = 2^{s+t} 3^{j-t} n + 2^t 3^{j-t} q - 1.$$

For every $t < j$, both terms preceding -1 are divisible by 3, and hence

$$Y_t(n) \equiv -1 \equiv 2 \pmod{3}.$$

Thus the next backward step is admissible. At $t = j$ we obtain

$$a_{3^j n + c} = a_{2^{s+j} n + r}, \quad r := 2^j q - 1.$$

It remains only to check the kernel range. Since $c \leq 3^j - 1$,

$$2^s c + 1 \leq 2^s (3^j - 1) + 1 \leq 2^s 3^j.$$

Hence $q \leq 2^s$. Therefore

$$0 \leq r = 2^j q - 1 < 2^{s+j}.$$

Thus $(a_{3^j n + c})_{n \geq 0}$ is a member of $K_2(a)$. □

Define the odd-indexed subsequence

$$b_n := a_{2n+1} \quad (n \geq 0). \quad (6)$$

Proposition 3.2. *If a is 2-automatic and satisfies (2), then b is both 2-automatic and 3-automatic.*

Proof. For every $e \geq 0$ and $0 \leq r < 2^e$,

$$(b_{2^e n + r})_{n \geq 0} = (a_{2^{e+1} n + (2r+1)})_{n \geq 0} \in K_2(a).$$

Hence

$$K_2(b) \subseteq K_2(a),$$

so b is 2-automatic.

Now let $e \geq 0$ and $0 \leq r < 3^e$. By (6),

$$b_{3^e n + r} = a_{2(3^e n + r) + 1} = a_{3^{e+1} n + (3r+2)}.$$

The constant $c = 3r + 2$ satisfies

$$0 < c < 3^{e+1}, \quad 3 \nmid c.$$

Lemma 3.1 therefore shows that every member of $K_3(b)$ lies in $K_2(a)$. Thus

$$K_3(b) \subseteq K_2(a).$$

Since $K_2(a)$ is finite, $K_3(b)$ is finite and b is 3-automatic. □

Corollary 3.3. *The sequence b is ultimately periodic.*

Proof. The integers 2 and 3 are multiplicatively independent. Proposition 3.2 and Cobham's theorem therefore imply that b is ultimately periodic. □

4 Elimination of the eventual period

The original invariance supplies two identities that will force the eventual period to collapse.

Lemma 4.1. *For every $n \geq 0$,*

$$b_{4n} = b_{3n}, \quad b_{4n+2} = b_n.$$

Proof. Using (2) and (6),

$$b_{4n} = a_{8n+1} = a_{12n+2} = a_{6n+1} = b_{3n},$$

and

$$b_{4n+2} = a_{8n+5} = a_{12n+8} = a_{6n+4} = a_{3n+2} = b_n.$$

□

Proposition 4.2. *If b is ultimately periodic and satisfies Lemma 4.1, then b is eventually constant.*

Proof. Choose N_0 and a least positive eventual period P such that

$$b_{n+P} = b_n \quad (n \geq N_0).$$

For $x \in \mathbb{Z}/P\mathbb{Z}$, define $f(x)$ as the common value of b_n for all sufficiently large $n \equiv x \pmod{P}$. This is well defined. Applying Lemma 4.1 to sufficiently large representatives of each residue class gives

$$f(4x) = f(3x), \quad f(4x+2) = f(x) \tag{7}$$

for every $x \in \mathbb{Z}/P\mathbb{Z}$. Notice that if $0 < d < P$ and $f(x+d) = f(x)$ for all x , then d is also an eventual period of b , contradicting the minimality of P .

First we show that P is odd. Put

$$d = \frac{P}{\gcd(P, 4)}.$$

Since $4d \equiv 0 \pmod{P}$, the second identity in (7) gives

$$f(x+d) = f(4x+4d+2) = f(4x+2) = f(x).$$

If $\gcd(P, 4) > 1$, then $0 < d < P$, contradicting the minimality of P . Thus $\gcd(P, 4) = 1$, so P is odd.

Next we show that $3 \nmid P$. Suppose instead that $3 \mid P$ and put $d = P/3$. For every x ,

$$f(4x+4d) = f(3x+3d) = f(3x) = f(4x),$$

where the first and third equalities use the first identity in (7). Since 4 is invertible modulo P , the residues $4x$ range over all of $\mathbb{Z}/P\mathbb{Z}$. Moreover $4d \equiv d \pmod{P}$ because $4d - d = 3d = P$. Hence $f(y+d) = f(y)$ for every residue y , again contradicting minimality. Therefore

$$\gcd(P, 6) = 1.$$

Because 3 and 4 are invertible modulo P , define affine permutations of $\mathbb{Z}/P\mathbb{Z}$ by

$$S(x) = \frac{3}{4}x, \quad F(x) = 4x + 2.$$

The first identity in (7), after replacing $4x$ by an arbitrary residue, says $f(Sx) = f(x)$; the second says $f(Fx) = f(x)$. Since S and F are permutations, f is invariant under their inverses and all compositions. A direct calculation gives

$$SFS^{-1}F^{-1}(x) = x - \frac{1}{2}.$$

Thus f is invariant under translation by $-1/2$. Since P is odd, 2 is invertible modulo P . Any integer representative u of $-1/2$ modulo P satisfies $\gcd(u, P) = 1$, so its additive order in $\mathbb{Z}/P\mathbb{Z}$ is

$$\frac{P}{\gcd(u, P)} = P.$$

Translation by u therefore acts transitively on $\mathbb{Z}/P\mathbb{Z}$. Hence f is constant. Consequently the least eventual period is $P = 1$, and b is eventually constant. $\square$

Lemma 4.3. *If b is eventually constant and $b_{4n+2} = b_n$ for all $n \geq 0$, then b is constant on $\mathbb{N}_0$.*

Proof. Let $F(n) = 4n + 2$. Iteration gives

$$F^k(n) = 4^k n + \frac{2(4^k - 1)}{3} \rightarrow \infty.$$

Since $b_{F(n)} = b_n$, we have $b_{F^k(n)} = b_n$ for every $k \geq 0$. For sufficiently large k , the index $F^k(n)$ lies in the constant tail, so b_n equals the tail value. This holds for every $n \geq 0$. $\square$

Proof of Theorem 1.1. By Proposition 3.2 and Cobham's theorem, b is ultimately periodic. Proposition 4.2 makes it eventually constant, and Lemma 4.3 makes it constant on all of $\mathbb{N}_0$; write $b_n = c$.

Every integer $N \geq 1$ has a unique representation

$$N = 2^s(2m + 1)$$

with $s, m \geq 0$. Repeated use of $a_{2n} = a_n$ therefore gives

$$a_N = a_{2m+1} = b_m = c.$$

Finally, $T(0) = 0$, and neither identity in (2) connects a_0 with a positive index. Moreover, for arbitrary choices $c, d \in \Delta$, the sequence

$$a_0 = d, \quad a_n = c \quad (n \geq 1)$$

is 2-automatic and satisfies (2). Thus a_0 is genuinely unconstrained. $\square$

5 Consequences for Collatz-invariant automatic sets

A subset $A \subseteq \mathbb{N}_0$ is called 2-automatic when its characteristic sequence is 2-automatic.

Corollary 5.1 (Invariant automatic sets are trivial on $\mathbb{N}_+$). *Let $A \subseteq \mathbb{N}_0$ be 2-automatic and suppose*

$$n \in A \iff T(n) \in A$$

for every $n \geq 0$. Then either

$$A \cap \mathbb{N}_+ = \emptyset$$

or

$$A \cap \mathbb{N}_+ = \mathbb{N}_+.$$

Membership of 0 is arbitrary.

Proof. Apply Theorem 1.1 to the characteristic sequence of A . □

Corollary 5.2 (Nontrivial invariant sets are not 2-automatic). *Let $A \subseteq \mathbb{N}_+$ satisfy*

$$n \in A \iff T(n) \in A \quad (n \geq 1).$$

If A is neither empty nor all of $\mathbb{N}_+$, then its characteristic sequence is not 2-automatic.

Proof. Regard $A \subseteq \mathbb{N}_+$ as a subset of $\mathbb{N}_0$, so its characteristic sequence satisfies $\chi_A(0) = 0$. Since $T(0) = 0$, the assumed invariance on positive integers extends trivially to $n = 0$. If χ_A were 2-automatic, Corollary 5.1 would imply that A is either empty or all of $\mathbb{N}_+$, a contradiction. □

Let

$$B := \{n \geq 1 : T^k(n) = 1 \text{ for some } k \geq 0\} \quad (8)$$

be the basin of 1. We regard its characteristic sequence as indexed by $\mathbb{N}_0$, with $0 \notin B$.

Lemma 5.3. *The set B satisfies*

$$n \in B \iff T(n) \in B \quad (n \geq 0).$$

Proof. For $n = 0$ both sides are false because $T(0) = 0$. Let $n \geq 1$. If $T(n) \in B$, then $T^k(T(n)) = 1$ for some $k \geq 0$, so $T^{k+1}(n) = 1$ and $n \in B$. Conversely, if $n \in B$ and $n \neq 1$, then a trajectory from n to 1 has positive length, so removing its first step shows $T(n) \in B$. For $n = 1$, one has $T(1) = 2$ and $T(2) = 1$, hence $T(1) \in B$ as well. □

Corollary 5.4 (Automaticity criterion for Collatz). *The Collatz conjecture on positive integers holds if and only if the characteristic sequence of B is 2-automatic.*

Proof. If the Collatz conjecture holds, then $B = \mathbb{N}_+$. Its characteristic sequence on $\mathbb{N}_0$ is 0 at 0 and 1 at every positive index, hence is 2-automatic.

Conversely, suppose the characteristic sequence of B is 2-automatic. Lemma 5.3 and Corollary 5.1 imply that B is either empty or all of $\mathbb{N}_+$. Since $1 \in B$, it follows that $B = \mathbb{N}_+$, which is the Collatz conjecture. □

Remark 5.5. Corollary 5.4 is a reformulation, not a proof, of the Collatz conjecture. The theorem does not establish that the basin indicator is 2-automatic. Instead, it says that if the Collatz conjecture is false, then the full basin of 1 must have genuinely non-2-automatic global complexity.

6 Relation to previous work and scope

The coefficient relations (2) are closely related to the functional-equation approach of Berg and Meinardus [3, 4], whose work studies analytic solution spaces associated with the Collatz recurrence. The present result instead concerns finite-alphabet sequences and automaticity.

Shallit and Wilson [12] proved automaticity results for predecessor families with a fixed number of odd Collatz steps. Colussi [7] gave a binary description of finite convergence classes, and Stérin [13] later constructed explicit regular expressions for ancestor sets with a bounded odd-step budget. These are finite-depth or bounded-budget inverse results. Theorem 1.1, by contrast, assumes complete invariance along every forward edge and proves that a global invariant coloring cannot be both nontrivial on the positive integers and 2-automatic.

Stérin and Woods [14] constructed a quasi-cellular automaton that exactly simulates the Collatz process and showed that its two-dimensional evolution embeds conversion from base 3

to base 2. Their result makes the interaction between these two bases explicit. The present theorem uses a different manifestation of the same base tension: Collatz invariance forces the 3-kernel of the odd-indexed subsequence into a finite 2-kernel, after which Cobham’s theorem applies.

Dhiman and Pandey [8] proved that, for generalized Collatz maps $T_{q,d}$ with q an odd prime and d odd, the arbitrary-step transition relation is not first-order definable in base-2 Büchi arithmetic. In a related paper [9], for a family including the classical map, they proved non-definability of the two-variable reachability relation in the corresponding Büchi arithmetic. Their arguments also exploit rigidity caused by multiplicatively independent bases, through Cobham–Seménov or Cobham-type obstructions. The objects, however, are different: those papers concern recognition of two-variable dynamical relations, whereas Theorem 1.1 concerns a one-variable finite coloring that is constant along each forward edge. Neither statement directly implies the other.

More generally, recognizability under affine closure and linear iteration has been studied by Klarner [10] and Boigelot [5]. Those results concern different classes of transformations and preservation questions and do not directly yield Theorem 1.1; they provide a broader automata-theoretic context for the rigidity phenomenon considered here.

Bell and Lagarias [2] studied generating functions of complete inverse orbits and obtained a natural-boundary/rationality dichotomy connected with the Collatz conjecture. Their analytic criterion and Corollary 5.4 are complementary in spirit: both isolate global structural complexity of inverse-orbit sets, but in different categories.

To the best of our knowledge, the specific statement proved here - that every finite-alphabet 2-automatic coloring invariant under the shortcut Collatz map is constant on the positive integers - has not previously been stated in this form. This is necessarily a limited literature claim rather than a proof of nonexistence of an equivalent result under different terminology.

7 Remarks and possible extensions

The proof has two logically distinct parts. First, Collatz invariance together with the fact that 2 generates the unit groups modulo 3^j forces

$$K_3(b) \subseteq K_2(a).$$

Second, the residual identities

$$b_{4n} = b_{3n}, \quad b_{4n+2} = b_n$$

destroy the eventual periodicity obtained from Cobham’s theorem. This separation suggests analogous rigidity problems for generalized piecewise-affine maps.

A natural problem is to characterize generalized Collatz maps for which nontrivial k -automatic invariant colorings exist. The present argument suggests that two ingredients may be decisive: a residue-class transfer mechanism that forces automaticity in a multiplicatively independent base, and enough residual affine symmetries to collapse the eventual period. Whether these conditions admit a useful general formulation remains open.

Formal verification note

The core proof chain has been formalized in Lean 4 with Mathlib. The formalization includes the arithmetic transfer lemma, the kernel inclusions, the period-collapse argument, the propagation from eventual constancy to global constancy of the odd-indexed subsequence, and bridges between finite k -kernels and standard most-significant-digit-first automaticity conventions,

including the treatment of leading zeros and the representation of 0. Cobham’s theorem itself is not formalized internally; it is supplied to the Lean development as an explicit external hypothesis. Thus the Lean development verifies the remaining proof chain conditional on that standard theorem. The Lean 4 source files used for the formal verification are publicly available in the accompanying GitHub repository.

Declaration on generative AI assistance

Generative AI assistance (OpenAI ChatGPT, GPT-5.6 Sol) was used during preparation of the manuscript for literature-search assistance, adversarial checking of proof steps, editorial drafting, and assistance with Lean 4 formalization and proof-code development. The AI system is not an author and is not used as a bibliographic source. Responsibility for verification of the mathematics, references, formalization, and final text rests with the human author.

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